

Chain of Thought as Repeated Measurement

Discrete and continuous reasoning are one Tensor Brain operator
at two settings of a collapse dial

Tensor Brain research note

Abstract

Chain-of-thought (CoT) reasoning emits a discrete token at every step; continuous latent reasoning, of which COCONUT [Hao et al., 2024] is the canonical instance, feeds a hidden state back instead and is reported to keep several candidate continuations alive at once. We observe that the Tensor Brain (TB) already contains both operations, and contains them as *one* operation. The TB measurement $q \leftarrow \alpha q + \beta a_k$ writes a single sampled index embedding back into the workspace; TB attention $q \leftarrow q + \beta \sum_k p_k a_k$ writes the expectation, and the QTB draft [Tresp and Li, 2025] already calls the latter a degenerate measurement whose outcome is not revealed. Discrete and continuous chains of thought are therefore the sharp and degenerate limits of a single update, and everything between them is reachable by turning one knob.

Making that identification lets us ask a question the LLM literature cannot easily pose, because in a transformer the token bottleneck is not adjustable: *when does measurement sharpness matter at all?* The TB retain gate α answers it. At $\alpha = 1$ the entire latent state survives the step and, we find, every collapse mode from a single sampled symbol to the exact expectation performs identically (spanning just 0.005 on our task). Sharpness only becomes a variable once α shrinks and the step boundary turns into a real channel. At $\alpha = 0$ the same conditions spread over 0.456–0.986, ordered exactly by how much the collapse operator can push through that channel. **Sharpness is not a property of a reasoning style; it is a property of a channel, and it only matters when there is one.** Narrowing the surviving state at $\alpha = 1$ partially reopens the gap, confirming that what matters is boundary capacity — but only partially, which is itself informative: a narrow continuous state and a sharp symbol are not interchangeable, because the first can still carry a superposition and the second cannot.

We then quantify the gap between the two limits. In unitary quantum computation measurements may always be deferred to the end of a circuit; in the TB the evolution operator is nonlinear, so measurement and evolution do not commute, and the discrepancy is exactly a Jensen gap. We prove it vanishes if and only if the evolution is affine, give its second-order form $\frac{1}{2}\beta^2 \text{tr}(\nabla^2 \Phi \text{Cov}_p[a])$, and confirm the predicted β^2 scaling to within 1.8 % numerically. The companion prediction — that the gap shrinks as the chain grows confident — holds when the candidate spread is swept directly, but is *flat* across the trained chain’s own steps, because on a search task the chain never becomes confident enough for the term to move. Collapse is never cheap here, which is why no annealed schedule we tried beats staying continuous. The same expansion separates two things the literature often conflates: averaging M measurements *inside* a step converges to continuous thought at rate $M^{-1/2}$, whereas self-consistency averages *across* whole chains and does not converge to it at all — the two differ by the accumulated Jensen gap. Finally, because the TB index layer exists at every step by construction, the superposition that latent reasoning is supposed to maintain can simply be read off rather than probed for, and repeated measurement of a single concept window — deliberation that costs no evolution steps — is shown to sharpen the index distribution *towards* the true search frontier.

All experiments are from-scratch models on a synthetic multi-hop reachability task and run in under an hour on a laptop CPU.

1 Introduction

A chain of thought is usually described as a sequence of intermediate results that a model writes down before answering [Wei et al., 2022]. Written down is the operative phrase: each step is forced through the vocabulary, so whatever the model was entertaining collapses to one token before the next step begins. COCONUT [Hao et al., 2024] removes that constraint by feeding the last hidden state back as the next input embedding, and reports that the resulting continuous thought behaves like a breadth-first search, keeping several candidate next steps alive where discrete decoding would have committed. The effect is large and it is selective: on ProsQA, a path-finding task over a DAG with distractor branches, CoT reaches 77.5% and COCONUT 97.0%, while on GSM8k arithmetic the ordering reverses, 42.9% for CoT against 34.1% for COCONUT.

The Tensor Brain [Tresp et al., 2021, Tresp and Li, 2025] describes cognition with two top-down operations that look, on inspection, like exactly these two regimes. A *measurement* samples a symbolic index k and writes its embedding back into the representation layer,

$$q \leftarrow \alpha q + \beta a_k, \quad k \sim p, \quad (1)$$

while *attention* writes the expected embedding without revealing any outcome,

$$q \leftarrow q + \beta \sum_k p_k a_k. \quad (2)$$

The QTB draft is explicit that (2) should be read as a degenerate measurement with unknown outcome, and separately proposes that repeated application of the evolution operator without fresh perceptual input is a form of chain-of-thought reasoning. Put those two remarks together and the claim writes itself: (1) and (2) are the same update with different feedback weights, discrete CoT is the sharp end and continuous thought the degenerate end, and the interpolation between them is a dial rather than a dichotomy.

This report takes that identification seriously and asks what it buys.

Contributions.

1. **One operator (§3).** We write token CoT, self-consistency, tempered decoding and continuous thought as a single TB update $q \leftarrow \alpha q + \beta \sum_k w_k a_k$ under different feedback weights w , and note that two of the four are already TB core operations.
2. **Sharpness is a property of the channel, not of the reasoning style (§3.1, §5.1).** The retain gate α controls how much of the latent state survives a step, and therefore whether the index layer is a channel at all. Sharpness is decisive at $\alpha = 0$ and irrelevant at $\alpha = 1$, and the transition is abrupt: retaining a quarter of the state shrinks the spread across all eleven collapse modes thirteenfold. This knob is not visible in a transformer, where the token bottleneck is fixed by the architecture, and the corresponding prediction — that any architecture carrying part of its reasoning state across the boundary is insensitive to how it decodes — is testable outside the Tensor Brain.
3. **Capacity and discreteness are correlated but not the same (§5.2).** Narrowing the state that survives at $\alpha = 1$ reopens the gap, so the effect is about what the boundary can carry. But the largest gap narrowing alone produces is far smaller than replacing the state with a symbol: a narrow continuous state still holds a superposition where a sharp one cannot. We therefore resist reducing the two knobs to a single scalar capacity.

4. **Deferred measurement fails, quantifiably (§3.2).** Measurement commutes with evolution iff the evolution is affine (Proposition 1); otherwise the gap is $\frac{1}{2}\beta^2 \text{tr}(\nabla^2\Phi \text{Cov}_p[a]) + O(\beta^3)$ (Proposition 2). The β^2 law is confirmed to within 1.8 %, with the expectation computed exactly rather than sampled. The confidence-dependence is confirmed only when the candidate spread is swept deliberately; across the trained chain’s own steps it is flat, because a search task never makes the chain confident. We report that as a negative result and draw the consequence for schedules from it.
5. **Inside-step averaging is not self-consistency (§3.3).** Averaging M draws within a step converges to continuous thought at $M^{-1/2}$; averaging across whole chains does not, and the difference is the accumulated Jensen gap. This gives a precise sense in which latent CoT is the mean-field limit of measurement, and self-consistency is not.
6. **Superposition you can read (§5.5).** Because the TB index layer is present at every step by construction, the search frontier is directly legible, with no Logit Lens or trained probe.
7. **Repeated measurement as deliberation (§3.4).** Measuring one concept window several times without evolving between measurements sharpens the index distribution towards the *true* frontier, at no cost in evolution steps, and has a stability threshold in (α, β) that we derive and observe.

What we do not claim. That continuous reasoning beats discrete reasoning on graph reachability is known, and Zhu et al. [2025b] prove a stronger version of it: a two-layer transformer with D steps of continuous CoT solves directed graph reachability where discrete CoT is only known to need $O(n^2)$ decoding steps, with the continuous thought carrying a superposition of search frontiers. Our reachability results should be read as a replication of that phenomenon inside a different, much smaller architecture, and the contribution claimed here is the measurement-theoretic framing, the role of α , and the quantities in items 3, 4 and 6 rather than the ordering itself.

2 Background and related work

Tensor Brain operations. We use the notation of the papers throughout: q is the representation-layer preactivation (pre-CBS), $\gamma = \sigma(q)$ is the cognitive brain state, A is the shared index bank whose column a_k is the embedding of symbolic index k , and candidate scores are $s_k = \gamma^\top a_k$ up to the score-mode offset. The evolution operator Φ moves the state between concept windows. Crucially the same A both scores indices and supplies the feedback, so a measurement outcome re-enters the workspace in the coordinates it was read out in.

Latent chain of thought. COCONUT [Hao et al., 2024] feeds the last hidden state back as the next input embedding, trained with a staged curriculum that progressively replaces written reasoning steps by continuous ones; without the curriculum it reports no gain over answering directly. Zhu et al. [2025b,a] give the theory and training dynamics of the superposition account. Soft Thinking [Zhang et al., 2025] and CoT2 [Gozeten et al., 2025] use probability-weighted mixtures of token embeddings as the continuous object, the former training-free; Butt et al. [2025] add noise to such mixtures for exploration and advocate training continuous and deploying discrete. Pause and filler tokens [Goyal et al., 2023, Pfau et al., 2024] add compute without adding content and are the right control for “more steps”; looped and recurrent-depth models [Dehghani et al., 2018, Geiping et al., 2025] instead keep the full state and iterate, which in our parameterisation is the $\alpha = 1$ corner.

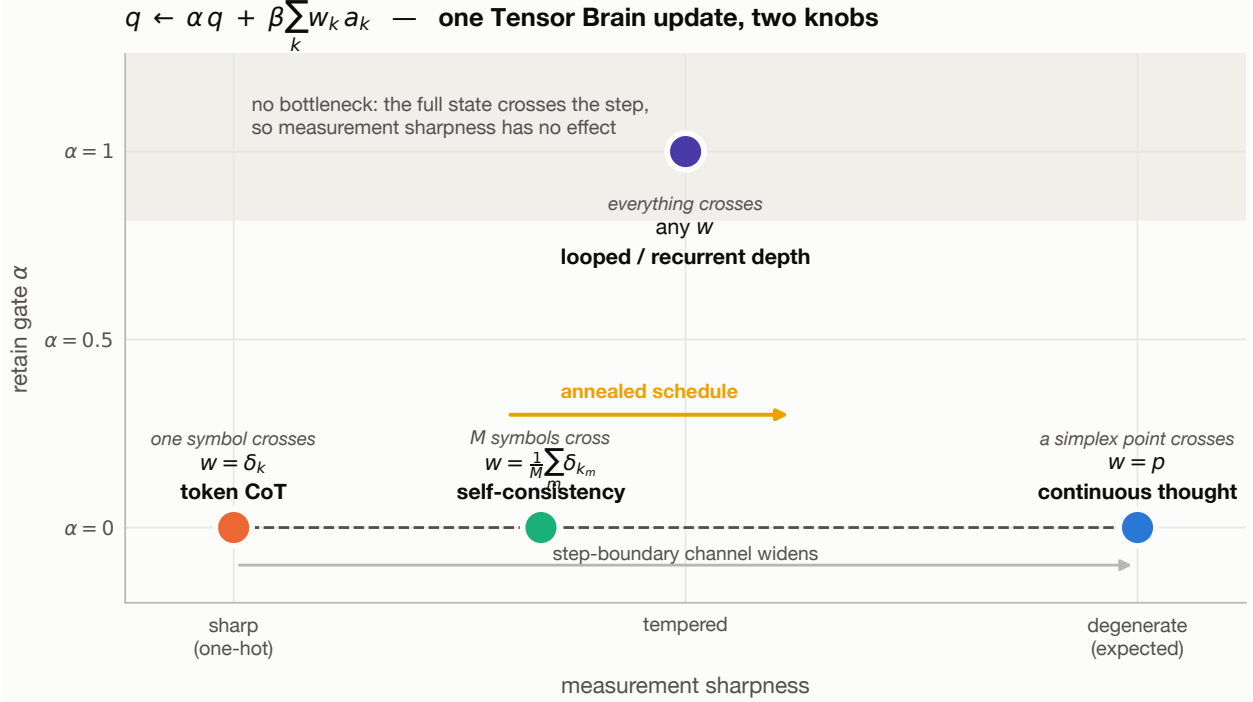


Figure 1: The two knobs of one Tensor Brain step update. The horizontal axis is measurement sharpness, which selects the feedback weights w and interpolates token CoT, self-consistency and continuous thought. The vertical axis is the retain gate α , which decides how much of the latent state survives the step and therefore whether the index layer is a bottleneck at all. In the shaded band the sharpness axis has no effect, because nothing is forced through the index layer.

Implicit CoT [Deng et al., 2023] and CCoT [Cheng and Van Durme, 2024] compress reasoning into hidden states by distillation.

Positioning. The ingredients of the dial exist. Mixture-of-embedding feedback is standard in the works above, and Zou et al. [2026] define a scalar “decisional certainty” governing an exploration–execution trade-off and argue that a curriculum is theoretically necessary — which matches what we find empirically in §4. What we have not found in the literature is the framing of discrete and continuous CoT as the sharp and degenerate limits of one measurement, the identification of the retain gate as the variable that decides whether sharpness matters at all, or the use of the deferred-measurement principle and its Jensen-gap failure as the quantity separating them. Quantum-cognition work in the Bussemeyer–Bruza tradition [Bussemeyer and Bruza, 2012] applies measurement formalism to human judgement, not to machine reasoning traces. Rizvi-Martel et al. [2026] caution that superposition is only actually used by models trained *from scratch* on latent thoughts, collapsing or being bypassed in training-free and fine-tuned regimes; our models are trained from scratch, which places this study squarely in the regime where the phenomenon is real, and equally means our results should not be read as claims about fine-tuned LLMs.

3 The collapse dial

Every intermediate reasoning step in this report applies

$$q \leftarrow \alpha q + \beta \sum_{k \in \mathcal{K}} w_k a_k, \quad p = \text{softmax}(s), \quad s_k = \gamma^\top a_k, \quad (3)$$

where $\mathcal{K}$ is the candidate set of the concept window. Conditions differ only in how w is formed:

Feedback weights w	Reads as	TB operation
$w = \delta_k, k \sim p$	token CoT, one collapse	measure, sample
$w = \delta_{\arg \max p}$	greedy CoT	measure, argmax
$w = \frac{1}{M} \sum_m \delta_{k_m}$	M measurements averaged	—
$w = \text{softmax}(s/\tau)$	tempered / partial collapse	—
$w = p$	continuous thought	attend
$w = 0$	no feedback	—
$w \mapsto u$ (fixed vector)	pause token	—

Two of the seven are TB core operations already, and the QTB reading of **attend** as a degenerate measurement is what makes the row ordering a continuum rather than a list. The temperature τ moves the *bias* of the estimator (how peaked the distribution being fed back is) and M moves its *variance* (how well a finite sample represents it); these are separate axes and we keep them separate throughout.

Remark 1 (What the unification does and does not assert). That attention and measurement share the form (3) is unconditional: both printed equations have it. Whether the two are *alternatives* is a separate question on which the two Tensor Brain papers disagree, and we do not resolve it here. Original TB Section 5.3 introduces the semantic-attention approximation as replacing the sampled-injection line, making attention an alternative to measurement. QTB instead states that Algorithm 2 (input and attention) and Algorithm 3 (generative measurement) execute *sequentially*, so that attention is part of input integration and precedes measurement rather than standing in for it. This report follows the original’s alternative reading, as the rest of the repository does, and its conditions should therefore be read as a controlled family of operators rather than as competing Tensor Brain schedules. A sequential condition, in which attention precedes measurement inside the same concept window, is the QTB-faithful companion and is not run here.

3.1 The step boundary as a channel

The knob that the LLM setting hides is α . In an autoregressive transformer, content generated at one step reaches the next only by being written as a token, and that is a property of the architecture, not a hyperparameter. Equation (3) exposes it. At $\alpha = 1$ the pre-CBS survives the step intact and the feedback is an *addition* to an otherwise complete state; at $\alpha = 0$ the state is *replaced* by the feedback, so everything the chain carries forward has to pass through $\sum_k w_k a_k$.

This makes the capacity of the step boundary explicit and orderable. A one-hot w transmits one symbol, at most $\log_2 |\mathcal{K}|$ bits. An M -sample average transmits a multiset of M symbols. A general $w = p$ transmits a point of the $(|\mathcal{K}| - 1)$ -simplex, which is real-valued and can represent a whole frontier at once. The prediction is immediate and is the main empirical claim of §5.1: *measurement sharpness should matter in proportion to how much the task needs to cross the boundary, and not at all when $\alpha = 1$.*

3.2 Deferred measurement and the Jensen gap

In unitary quantum computation the deferred-measurement principle says intermediate measurements can always be postponed to the end of the circuit without changing the output distribution [Nielsen and Chuang, 2010]. It holds because the evolution is linear. The TB evolution operator is not, and this is exactly where discrete and continuous chains part company.

Write $\bar{a} = \sum_k p_k a_k$ and $\bar{q} = \alpha q + \beta \bar{a}$ for the state a degenerate measurement produces, and $q_k = \alpha q + \beta a_k$ for the state outcome k produces. Deferring the measurement means evolving $\bar{q}$; performing it means evolving q_k and averaging over k .

Proposition 1 (When measurement may be deferred). *Let Φ be continuous on the convex hull C of $\{q_k\}_{k \in \mathcal{K}}$. Then $\Phi(\bar{q}) = \mathbb{E}_{k \sim p}[\Phi(q_k)]$ for every distribution p on $\mathcal{K}$ and every q if and only if Φ is affine on C .*

Proof. If Φ is affine the two sides agree by linearity of expectation. Conversely, taking $p = \frac{1}{2}(\delta_j + \delta_k)$ gives $\Phi(\frac{x+y}{2}) = \frac{1}{2}(\Phi(x) + \Phi(y))$ for all $x, y \in \{q_k\}$, and applying the same identity to the resulting midpoints extends it to all dyadic convex combinations, which are dense in C . A continuous solution of Jensen’s functional equation on a convex set is affine. $\square$

So the two ends of the dial compute the same thing only in the degenerate case where the evolution operator does no nonlinear work — that is, only when the chain is not reasoning. The size of the discrepancy is second order in the feedback gate.

Proposition 2 (Second-order gap). *If Φ is twice continuously differentiable at $\bar{q}$, then*

$$\mathbb{E}_{k \sim p}[\Phi(q_k)] - \Phi(\bar{q}) = \frac{1}{2} \beta^2 \left(\text{tr}(\nabla^2 \Phi_j(\bar{q}) \text{Cov}_p[a]) \right)_j + O(\beta^3), \quad (4)$$

where Φ_j is the j -th output coordinate and $\text{Cov}_p[a] = \sum_k p_k a_k a_k^\top - \bar{a} \bar{a}^\top$.

Proof. Put $\delta_k = a_k - \bar{a}$, so $q_k = \bar{q} + \beta \delta_k$ and $\mathbb{E}_p[\delta_k] = 0$. A second-order Taylor expansion of each output coordinate about $\bar{q}$ gives $\Phi_j(q_k) = \Phi_j(\bar{q}) + \beta \nabla \Phi_j(\bar{q})^\top \delta_k + \frac{1}{2} \beta^2 \delta_k^\top \nabla^2 \Phi_j(\bar{q}) \delta_k + O(\beta^3)$. Taking expectations kills the linear term, and $\mathbb{E}_p[\delta_k^\top H \delta_k] = \text{tr}(H \text{Cov}_p[a])$. $\square$

Two consequences do the work later.

Corollary 1 (Confidence is free, hedging is not). *The leading term of (4) vanishes iff $\text{Cov}_p[a] = 0$ on the span of the candidate embeddings, i.e. iff p is a point mass. A chain that is already certain loses nothing by collapsing; a chain that is hedging loses precisely the curvature-weighted spread of what it was hedging over.*

Corollary 1 is the principled version of a heuristic that appears repeatedly in the latent-reasoning literature — commit when confident, stay continuous when uncertain — and it is what motivates the annealed schedules of §5.7. It also predicts the shape of Figure 5(b): the gap should grow with the index entropy of the step, because entropy and $\text{tr Cov}_p[a]$ move together for a near-orthonormal index bank.

3.3 Averaging inside a step is not self-consistency

Self-consistency [Wang et al., 2022] samples whole chains and marginalises their answers. It is tempting to read continuous thought as the limit of that procedure. It is not, and Proposition 2 says why.

With M i.i.d. draws the feedback is $\hat{f}_M = \frac{1}{M} \sum_m a_{k_m}$, an unbiased estimator of $\bar{a}$ with covariance $\frac{1}{M} \text{Cov}_p[a]$, so

$$\mathbb{E} \|\hat{f}_M - \bar{a}\| = \Theta(M^{-1/2}) \sqrt{\text{tr Cov}_p[a]}. \quad (5)$$

Averaging *inside* a step therefore converges to the continuous update at rate $M^{-1/2}$: widen the channel enough and a chain of measurements becomes a chain of continuous thoughts. Averaging *across* chains does something different. Let $\Phi^{(H)}$ denote H composed steps; self-consistency estimates $\mathbb{E}[\Phi^{(H)}(\cdot)]$ over sampled trajectories, whereas continuous thought computes $\Phi^{(H)}(\mathbb{E}[\cdot])$ step by step. These differ by the Jensen gap accumulated at every step, and no number of samples closes it. In the language of §3.2, self-consistency defers nothing and averages at the end; continuous thought defers everything and never averages.

3.4 Repeated measurement of one window

The QTB draft raises repeated operator application without fresh input as a model of deliberation. The dial makes a sharper version available: measure the *same* concept window several times, with no evolution in between. Writing R for the number of repeats, the iteration is

$$q_{r+1} = \alpha q_r + \beta \sum_k w_k(q_r) a_k, \quad (6)$$

which is a fixed-point map on a single window. Because feedback re-enters through the same A that produced the scores, measuring an index raises that index’s score, and the window self-reinforces. This is the mechanism behind the quantum Zeno effect [Misra and Sudarshan, 1977], where frequent measurement freezes a state, transplanted to a setting where the “state” is a belief over symbolic indices.

Linearising $\sigma(q) - \frac{1}{2} \approx q/4$ and taking a near-orthonormal index bank, $A^\top A \approx I$, the score iteration becomes $s \leftarrow \alpha s + \frac{\beta}{4} \text{softmax}(s)$. The maximally uncertain state $s \propto \mathbf{1}$ is a fixed point, and the Jacobian there is $\alpha I + \frac{\beta}{4}(\text{diag}(p) - pp^\top)$ with eigenvalue $\alpha + \frac{\beta}{4K}$ on the subspace orthogonal to $\mathbf{1}$, where $K = |\mathcal{K}|$.

Proposition 3 (Commitment threshold, small-signal regime). *In this linearisation the uncertain fixed point of (6) is stable iff $\alpha + \beta/(4K) < 1$; above that threshold repeated measurement drives the window towards an absorbing one-hot state with no evolution step. For $\alpha < 1$ the non-uniform fixed points satisfy $s^* = \frac{\beta}{4(1-\alpha)} p(s^*)$, so $\beta/(1-\alpha)$ sets how peaked the window ends up.*

Remark 2 (This threshold does not survive contact with the trained model). We state Proposition 3 because it is the natural first analysis and because it is *wrong* about the regime we actually measure, in an instructive way. With $K = 64$ it predicts that $\alpha = 0.5$ sits far inside the stable region for every gate we sweep, so a half-decaying window should not sharpen at all. Section 5.8 finds that $\alpha = 0.5$ sharpens *fastest*. The linearisation is at fault: it expands σ about the origin, whereas a trained chain writes states whose norm puts σ deep into saturation, and the fixed-point scale $\beta/(1-\alpha)$ — which grows as α falls — dominates the local stability calculation. The small-signal threshold is therefore not a prediction about trained Tensor Brains, and we report the repeated-measurement behaviour empirically instead.

Whether the resulting commitment is *useful* — whether the window sharpens towards the correct answer or merely towards its own first guess — is not settled by either calculation, and §5.8 answers it directly.

4 Task and setup

Task. We use a TB-native analogue of ProsQA: a fixed layered DAG over concept nodes, with 400 start concepts and four subsequent layers of 64 nodes each, branching factor 2. A query writes a start concept and two candidate terminal concepts into the workspace; exactly one terminal is reachable, and the model must say which. The graph is fixed across the whole dataset, so it is knowledge that has to live in the model rather than in the input.

Removing the shortcuts. Two controls make the task about traversal rather than association, and both were necessary — earlier versions of this study produced uniformly high accuracy for every condition until they were added.

- *Quota-balanced negatives.* A terminal that many starts reach is, for that reason, rarely available as somebody else’s unreachable option, so matching negatives on in-reachability count is not enough. Negatives are assigned under a quota making each terminal occupy the correct and incorrect slot equally often. A terminal-identity rule fitted on the training split then scores 0.45 on the held-out split, i.e. noise.
- *Frozen index bank.* A trainable A of realistic size simply stores (start, terminal) associations and answers without traversing, at which point every collapse condition scores identically. We therefore freeze A to a fixed random code. All task knowledge lives in the evolution operator, and superposition readout is the only route to the answer.

Splits are over distinct reachable (start, terminal) pairs, so every start and every terminal is seen in training but their composition is not.

Supervision and curriculum. Intermediate steps here are inherently multi-valued: several children of the current frontier are equally valid continuations, so a one-hot “gold path” target is misspecified and teaches an arbitrary tie-break. We supervise instead towards the true breadth-first *frontier* at each hop. Training follows COCONUT’s staged curriculum: at stage k the first k intermediate hops use the condition’s own collapse and the rest are teacher-forced onto the true frontier, with step loss only at teacher-forced hops. The teacher is a *distribution*, not a node: writing back a single gold node would put one path in the workspace while the step target is the whole frontier, a target that state cannot support, and getting this wrong was what kept an earlier version of the chain from learning at all. Teacher forcing is training-only and never an evaluation condition, since writing the true frontier into the workspace leaks the answer. Reaching any nontrivial accuracy required this curriculum, independently matching the necessity argument of Zou et al. [2026].

Model and controls. A single shared evolution operator (QTB Algorithm 1, one sigmoid hidden layer of 1024 units on a 384-dimensional pre-CBS) is applied at every hop; the candidate set at hop h is layer h , identically across conditions. Discrete modes use a straight-through estimator [Bengio et al., 2013, Jang et al., 2017] with the tempered softmax as surrogate. Two controls separate “more compute” from “more information”: *no feedback* ($w = 0$) and a *pause vector* (a learned direction independent of the candidate distribution), the analogue of pause tokens [Goyal et al., 2023].

A confound worth naming. A flat w averages many near-orthogonal embeddings towards zero, so soft feedback is intrinsically smaller in norm than a one-hot collapse — in our setting by roughly $9\times$. Comparing them naively confounds measurement sharpness with how hard the step pushes

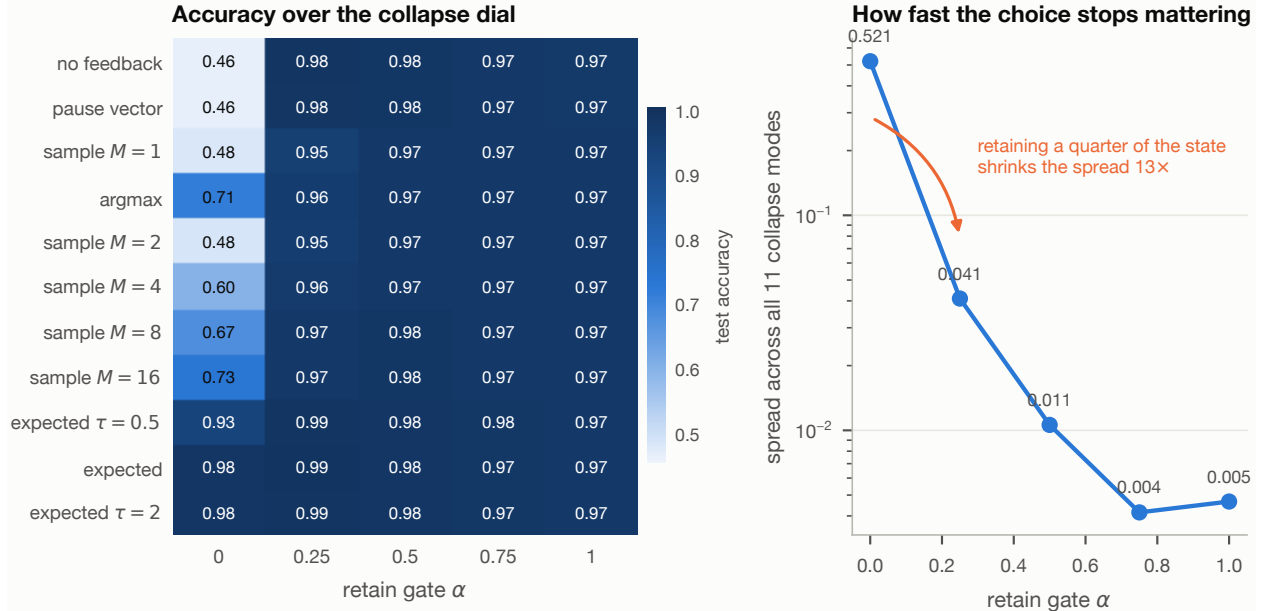


Figure 2: Test accuracy over the collapse dial, 3 seeds, 3258 held-out queries. **Left:** the full grid of retain gate against collapse mode. Only the $\alpha = 0$ column varies; everywhere else the choice of collapse is worth less than a percentage point. **Right:** the spread across all eleven modes as a function of α , on a log axis. The dependence is steep rather than gradual — retaining a quarter of the state already removes most of the effect.

on the state. We rescale all feedback to a common norm before applying β , so that the dial varies direction and not magnitude.

5 Results

5.1 Sharpness matters only at the bottleneck

Figure 2 is the central result. At $\alpha = 1$ every condition lands in a band 0.005 wide — including *no feedback*, which never touches the index layer at all. Whatever the chain needs is already carried by the pre-CBS, so the measurement is decorative and its sharpness is not a variable.

At $\alpha = 0$ the same eleven conditions spread over 0.521, and they order themselves by how much the collapse can transmit and by nothing else. The no-feedback and pause controls sit at chance (0.461 and 0.460); a single sampled symbol is barely above it (0.479); the M -sample average climbs monotonically with M (0.479, 0.482, 0.595, 0.674, 0.730 for $M = 1, 2, 4, 8, 16$); and the exact expectation reaches 0.981. The temperature axis moves the same way in isolation: $\tau = 2$ and $\tau = 1$ are indistinguishable at 0.979 and 0.981, $\tau = 0.5$ falls to 0.926, and the hard limit reaches 0.713. That the pause control fails while the same number of evolution steps with informative feedback succeeds is what rules out “extra compute” as the explanation, exactly as pause tokens do in the LLM setting.

Two details are worth more than the ordering itself.

The transition is abrupt, not gradual. We expected the spread to close smoothly as α rose. It does not. Retaining a mere quarter of the pre-CBS shrinks the spread across all eleven modes from 0.521 to 0.041, a factor of thirteen, and by $\alpha = 0.75$ it is under 0.005 — a factor of 130. Whatever the step boundary is doing, it is close to an on/off switch rather than a dial in this dimension,

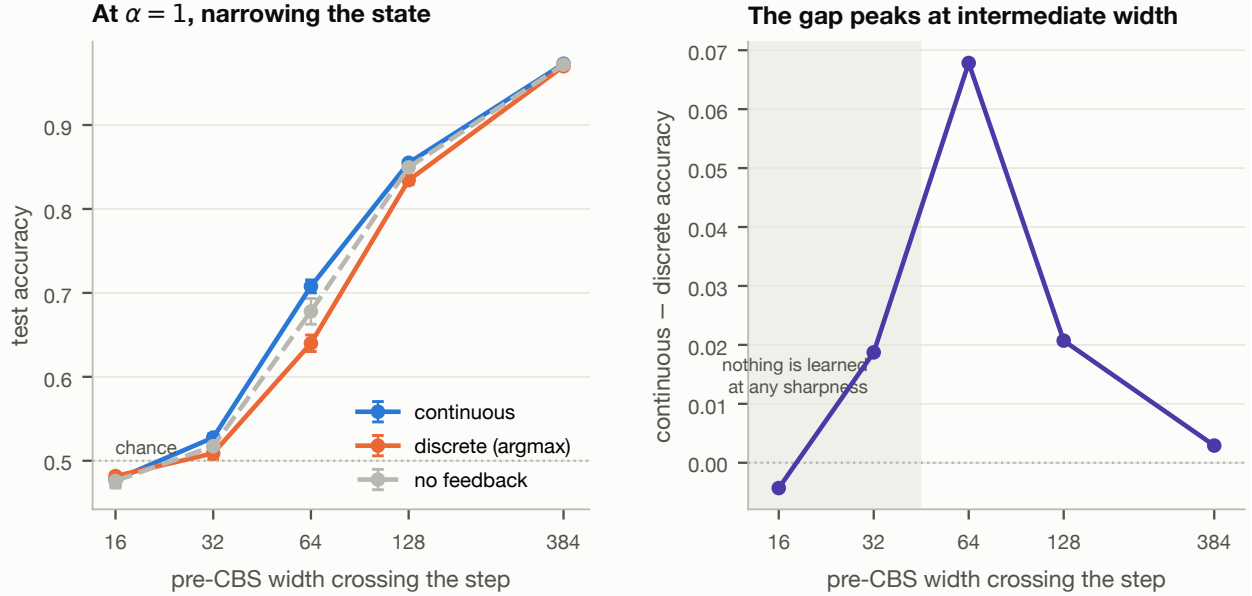


Figure 3: Holding $\alpha = 1$ and narrowing the pre-CBS that crosses the step. **Left:** the three headline conditions against state width. **Right:** the continuous minus discrete accuracy gap. If the $\alpha = 1$ null were about the retain gate as such it should persist at every width; if it is about how much information the boundary can carry, the conditions should separate again once the state is too narrow to hold the frontier.

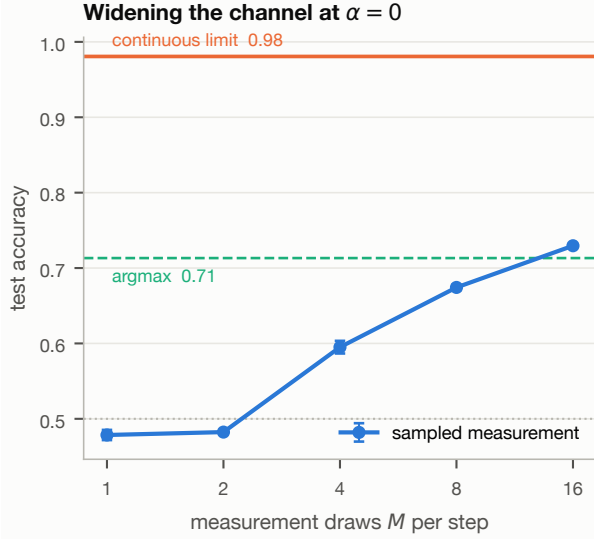
even though the sharpness axis *is* a smooth dial at $\alpha = 0$. The practical form of this claim is falsifiable outside the Tensor Brain: any architecture that carries even a fraction of its reasoning state across the step boundary — a residual path, a persistent cache, a recurrent hidden state — should be correspondingly insensitive to whether its intermediate steps are decoded discretely, and the discrete-versus-continuous question should only bite where the boundary is close to total.

Feedback can be worse than none. At $\alpha = 0.25$ the best condition is *no feedback* at 0.983, ahead of a single sampled measurement at 0.947. Once the retained state already carries the frontier, writing a sampled index back into it adds noise to a sufficient statistic. Measurement is not free even when it is cheap, which is a second reason not to read the collapse dial as “more symbolic grounding is better”.

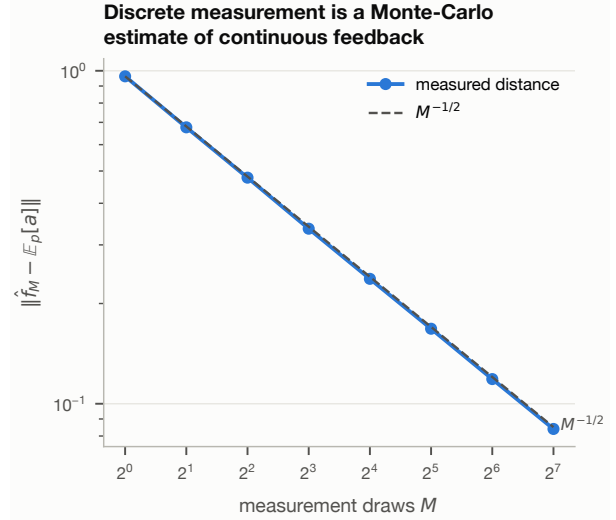
5.2 Is it the gate, or the capacity?

Section 5.1 is easy to over-read. What crosses the boundary at $\alpha = 1$ is a 384-dimensional state, comfortably wider than the frontier this task needs; what crosses at $\alpha = 0$ under a one-hot collapse is about $\log_2 64 = 6$ bits. Both knobs may therefore be moving one underlying quantity, the capacity of the step boundary, with α merely the coarser of the two. Figure 3 tests this directly by holding $\alpha = 1$ and shrinking the pre-CBS.

The answer is a qualified yes, and the qualification is the interesting part. The gap between continuous and discrete feedback traces an inverted U in state width. It is 0.003 at 384 dimensions, rises to 0.068 at 64 dimensions — roughly twenty times larger — and falls back to nothing at 16–32 dimensions, where no condition exceeds 0.482 and the model simply fails to learn the task at all. Read past that floor, the middle of the curve is a genuine confirmation: the $\alpha = 1$ null is not a property of the retain gate, and narrowing what survives the step brings the collapse modes apart again.



(a) Accuracy against draws per step.



(b) Feedback distance to the exact expectation.

Figure 4: Averaging measurements inside a step. (b) confirms the $M^{-1/2}$ rate of (5) with the expectation computed exactly over the candidate set.

What the sweep also shows is that the two axes are *not* interchangeable. The largest gap narrowing alone can produce is 0.068, while replacing the state with a single symbol at $\alpha = 0$ produces 0.277. A 64-dimensional continuous state and a 6-bit symbol are both small, but the first still holds graded evidence about many candidates at once and the second holds one node. Capacity in bits is the wrong summary; what the boundary must be able to carry is a *superposition*, and dimension buys that where discreteness forecloses it. We would therefore resist collapsing the two knobs of Figure 1 into a single scalar, and read Figure 3 as showing that they are correlated rather than the same.

5.3 Widening the channel

Figure 4 tracks the estimator directly, and (5) is met almost exactly. Going from $M = 1$ to $M = 128$ draws, the distance between the sampled feedback and the exact expected feedback falls from 0.96 to 0.084, a ratio of 11.5 against the $\sqrt{128} = 11.3$ the rate predicts. Accuracy rises with M in step, from 0.479 at a single draw to 0.730 at sixteen, against 0.981 for the exact expectation.

This is the sense in which continuous thought is the mean-field limit of repeated measurement. It is worth being precise about what it is *not*. Self-consistency [Wang et al., 2022] also averages many sampled chains, and it is tempting to read latent reasoning as its limit; §3.3 says otherwise. Averaging inside a step converges here at $M^{-1/2}$ because the object being averaged is the feedback vector, which enters the next evolution linearly. Averaging across whole trajectories composes the nonlinear Φ before averaging, and by Proposition 1 that limit differs from the continuous chain by the Jensen gap accumulated at every step — no number of samples closes it. The two procedures look alike and converge to different things.

5.4 Measurement does not commute with evolution

One scope note. Propositions 1 and 2 are statements about the printed update $q \leftarrow \alpha q + \beta \sum_k w_k a_k$, and that is the operator measured here, evaluated at states a trained chain actually visits. The

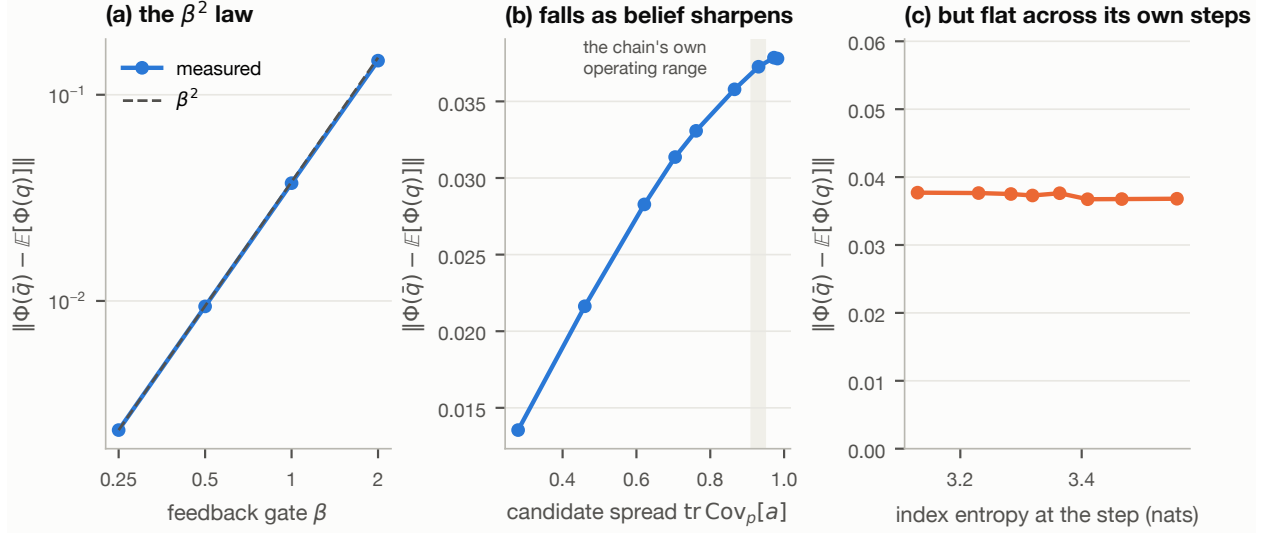


Figure 5: The Jensen gap between evolving the averaged state and averaging the evolved states, computed exactly over the candidate set. **(a)** the predicted β^2 law of Proposition 2. **(b)** tempering the chain’s own scores sweeps the candidate spread across its full range; the gap falls steeply towards zero as p sharpens, in the direction Corollary 1 requires, and the shaded band marks where the trained chain’s own steps actually sit. **(c)** bucketing those steps by entropy instead gives a flat line — not a refutation, but a range problem, since the spread is saturated everywhere in (b)’s shaded band.

chain *conditions* instead apply the norm-matched variant of §4, so that the dial varies direction rather than magnitude. The two coincide up to the per-step rescaling, and the β sweep below should be read as characterising the operator, not as a sweep of the trained model’s own gate.

The β^2 law of Proposition 2 holds almost exactly: across the swept gates the measured gap ratios deviate from the predicted factor of four by at most 1.8 %. This is the quantitative content of the claim that latent and token reasoning are not two implementations of the same computation — the difference is a number, and it is computable at every step.

Corollary 1 fares less simply, and the way it fails is informative. Bucketing the trained chain’s own steps by index entropy gives an essentially *flat* gap (Figure 5c): entropy 3.13 to 3.56 nats moves it by a ratio of 0.98. That is not a refutation of the corollary, which predicts the gap vanishes when $\text{Cov}_p[a] \rightarrow 0$, i.e. at a *point mass* — and on a search task the chain is never anywhere near one. Its candidate spread sits at 0.91–0.95 of the maximum at every step, so entropy varies while the quantity the theory actually names does not, and the cross-section has no range to test over. Driving the spread across its full range by tempering the chain’s own scores (Figure 5b) recovers the predicted behaviour: the gap collapses towards zero as p sharpens.

The practical reading is worth stating plainly, because it explains a result three sections below. On this task collapse is *never* cheap: the chain never becomes confident enough for the second-order term to shrink, so there is no step at which committing is close to free. That is exactly why no annealed schedule in §5.7 beats staying continuous throughout, and it predicts that “commit when confident” will only pay on tasks whose intermediate steps actually become confident — arithmetic, plausibly, rather than search.

5.5 Reading the superposition off the index layer

The TB gives this analysis for free. In an LLM the continuous thought has to be decoded with a Logit Lens or a trained probe to see what it contains; in the TB the index layer is present at every

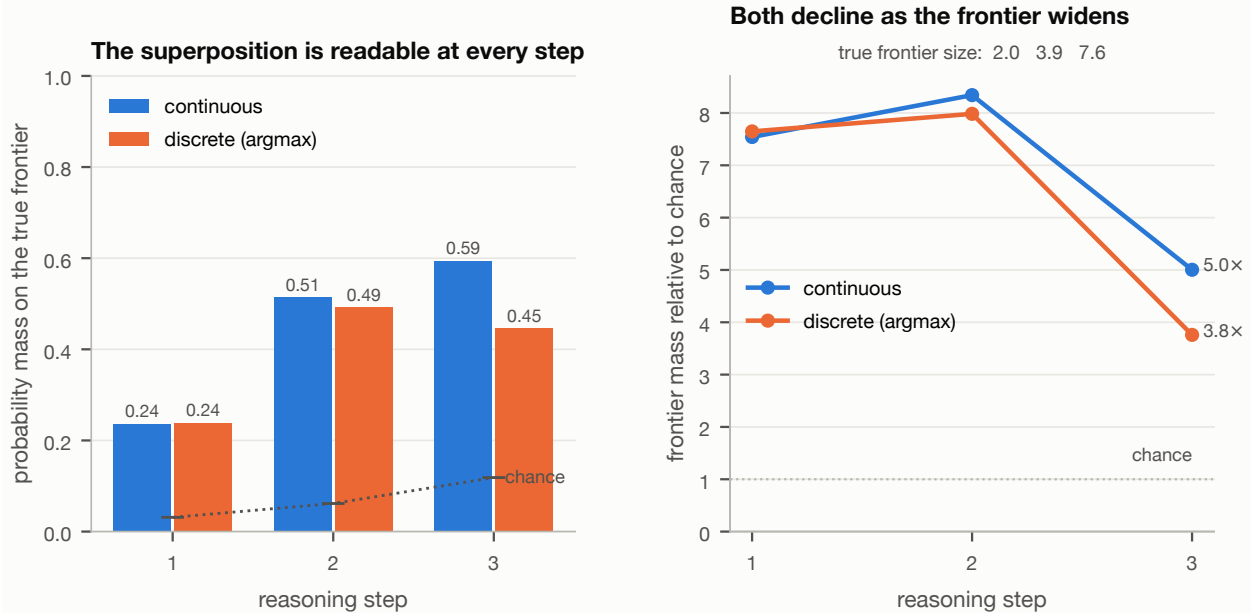


Figure 6: Left: probability mass on the true breadth-first frontier at each step, for a continuous and a discrete chain. **Right:** effective number of alternatives e^H kept open, against the true frontier size.

step by construction, so the candidate distribution *is* the readout, with no probe to validate and nothing to trust on faith.

What it shows is more specific than “continuous chains represent more”. Both chains put far more mass on the true frontier than chance (0.070), and at the first two steps they are close to indistinguishable — 0.236 against 0.239 at step one, 0.514 against 0.492 at step two. They separate only at step three, where the frontier is widest, at 0.593 against 0.446. Yet the accuracy difference between these two chains is enormous: 0.45 against 0.39 mean frontier mass corresponds to 0.983 against 0.706 test accuracy.

That mismatch is the point, and it sharpens the interpretation of §5.1. The discrete chain is not much worse at *knowing* where the search could be; its index distribution is nearly as informative. It is worse at *transmitting* that knowledge across the step boundary, because a one-hot collapse can forward only one node of a frontier that by step three averages 7.6. The deficit created by measuring sharply is a communication deficit, not a representational one — which is exactly why widening the channel in §5.3 recovers the accuracy without changing anything about what the chain believes.

5.6 Search depth

If the deficit created by sharp measurement is a communication deficit, it should scale with how much there is to communicate. Figure 7 varies the search depth from 2 to 5 hops, which grows the terminal frontier from 4 nodes to 21.

The continuous chain is essentially *depth-invariant*: it scores between 0.947 and 0.981 at every depth, because carrying a frontier of twenty-one costs it no more than carrying a frontier of four. Every discrete condition decays monotonically, and the gap grows from 0.143 to 0.386 — close to linear in frontier size over this range.

The M -sample condition dates the mechanism precisely. An M -draw collapse forwards at most M distinct nodes, so it should track the continuous chain while M exceeds the frontier and fall away once it does not. At $M = 8$ that crossing should happen between three and four hops, and it

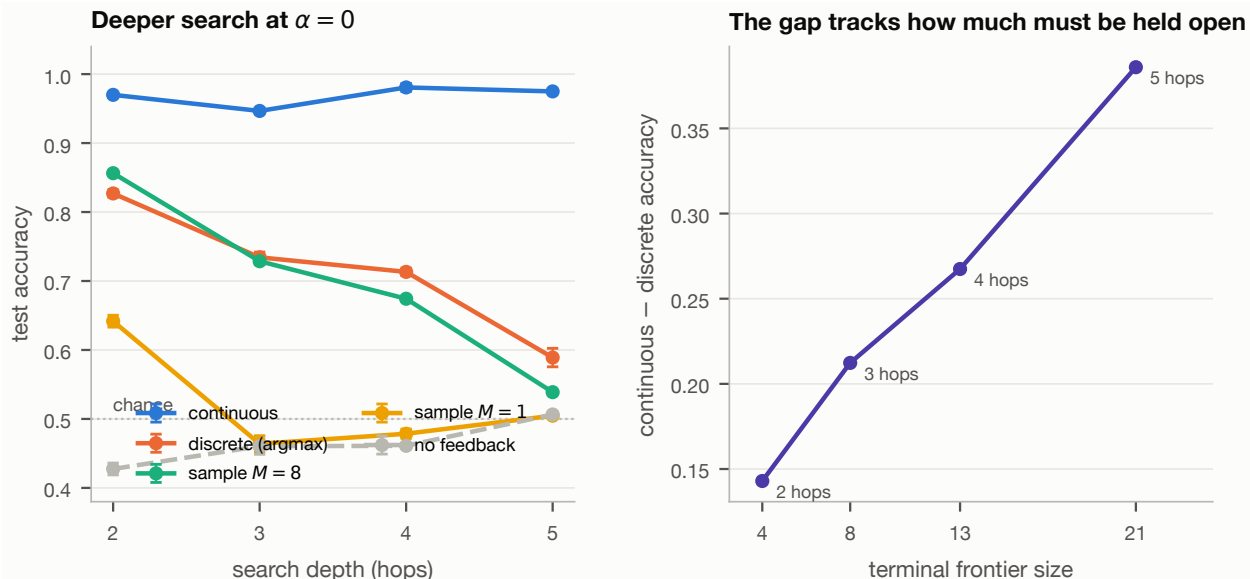


Figure 7: Search depth at $\alpha = 0$, three seeds. **Left:** continuous feedback is flat in depth while every discrete condition decays. **Right:** the gap against the size of the terminal frontier the chain has to represent.

does: 0.856 at a frontier of four, then 0.539 at a frontier of twenty-one. This is the same M axis as §5.3, now with the frontier rather than the sample count as the moving part, and it agrees.

5.7 Where in the chain to collapse

Corollary 1 says collapse is free when the step is confident and expensive when it is not. The frontier widens with depth on this task, so the prediction is that *where* a chain collapses should matter as much as whether it does, and that collapsing early should cost less than collapsing late.

Figure 8 bears this out sharply. Among the three schedules that collapse exactly once, accuracy is monotone in the position of that collapse: 0.792 at the first step, 0.755 at the second, 0.669 at the third, against 0.981 for a chain that never collapses. One collapse step is not one cost; it is a cost that scales with how much was being held open when it landed.

The comparison we did not anticipate is more interesting. Collapsing at *every* step (0.713) is *better* than collapsing only at the last one (0.669). A chain that measures throughout is trained to work from single nodes and never builds a superposition it cannot transmit; a chain that stays continuous and then collapses immediately before readout spends three steps assembling a frontier and discards it at the one moment it was needed. Consistency between what the chain represents and what the step boundary can carry is worth more than continuity at any individual step, which is a caution against bolting a discrete decoding step onto the end of an otherwise latent reasoner.

5.8 Repeated measurement of one window

This is the operation the QTB draft gestures at, and on this task it works. Holding one concept window fixed and measuring it 12 times with no evolution and no new perceptual input, the index entropy falls from 3.34 to 1.65 nats, and — the part that matters — probability mass on the true frontier rises from 0.510 to 0.819. The window does not merely become more certain; it becomes more *right*, at the cost of no evolution steps whatsoever.

Two controls establish that this is deliberation rather than a self-fulfilling loop. First, agreement

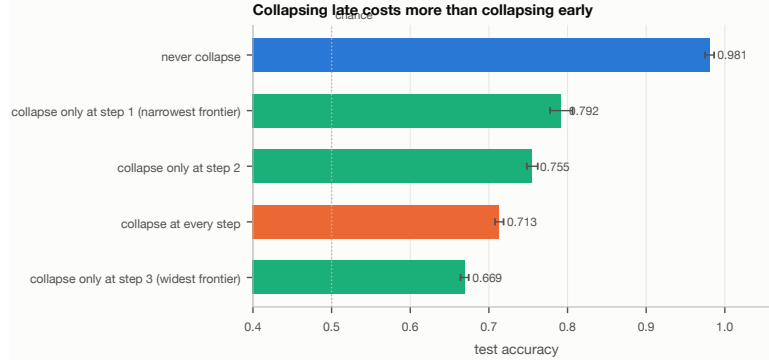


Figure 8: Where a chain collapses, at $\alpha = 0$. The three middle bars each collapse exactly once, at the first, second and third reasoning step respectively; the frontier widens with depth, so the same single collapse costs more the later it lands. Collapsing at every step beats collapsing only at the last.

with the window’s own first outcome falls to roughly 0.57 and plateaus (Figure 9c): in over forty percent of queries the window ends up committed to a different index than the one it would have named initially, so the feedback is not simply amplifying whatever it happened to see first. Second, the improvement is not automatic. At $\alpha = \beta = 0.5$ — weak feedback written into a state that halves every repeat — frontier mass *falls* from 0.51 to 0.33 before partially recovering. Repeated measurement helps only when what it writes back accumulates faster than the retained state decays.

We flag the obvious caveat: these trajectories are read from a chain trained with a single measurement per window, so they show what repeated measurement does to a representation that was not optimised for it. Training under a repeated-measurement schedule is the natural follow-up and is not run here.

6 Discussion

The question is about the channel, not the representation. The result we would most like to carry over is the α finding. Discussions of latent reasoning usually compare a discrete pipeline against a continuous one and attribute the difference to the representation. Equation (3) says the representation matters only in proportion to how much of the state is destroyed at the step boundary, and that the dependence is steep: a quarter of the state retained is already most of the way to not mattering. For transformers this suggests the useful question is not “discrete or continuous?” but “how much of the reasoning state actually has to survive re-entry?”, and architectures differ in that far more than in the sharpness of their decoding. It also suggests a cheap diagnostic. Before asking whether a system would benefit from latent reasoning, measure how much of its reasoning state already crosses its step boundary; if the answer is “most of it”, the continuous variant has little room to help, whatever the benchmark says.

Two knobs, not one. We were tempted to collapse α and sharpness into a single scalar capacity, and §5.2 says not to. Narrowing a continuous state and discretising it are both ways of shrinking what crosses the boundary, but they are not equivalent at equal nominal size, because only one of them can still carry a superposition. Whatever the right currency for step-boundary capacity is, it is not bits.

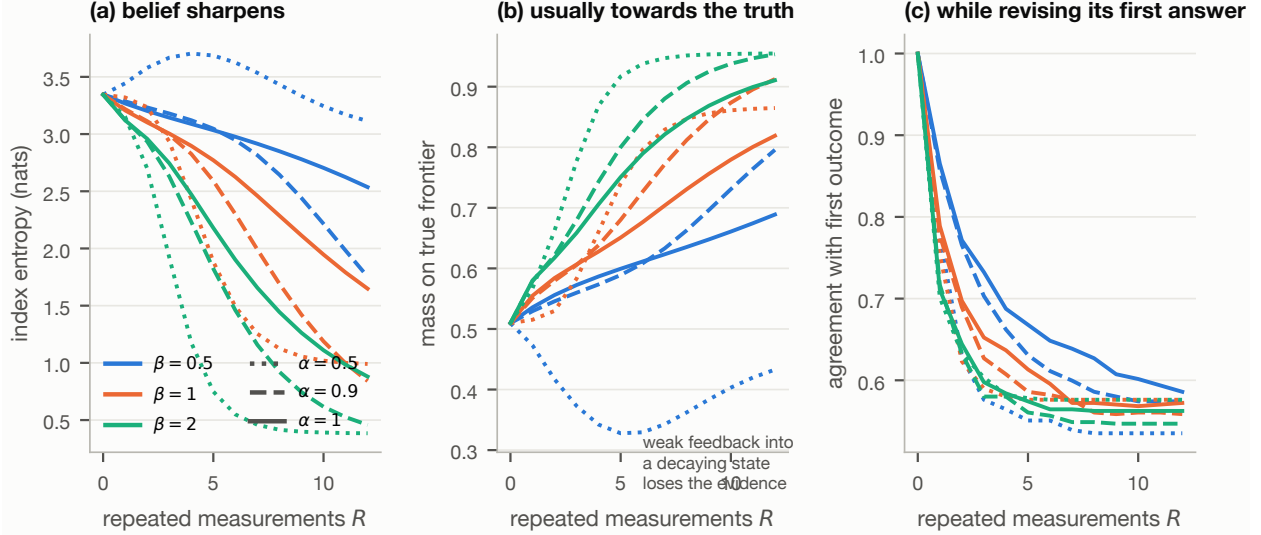


Figure 9: Repeated measurement of a single concept window, with no evolution and no new input in between, across nine (α, β) settings. (a) the index distribution sharpens in every setting, including ones Proposition 3 calls stable. (b) in eight of nine settings the sharpening moves *towards* the true frontier; the exception is weak feedback into a decaying state, where the evidence washes out faster than it accumulates. (c) agreement with the window’s own first outcome falls to about 0.57 and stays there, so this is revision rather than self-reinforcement.

Deferred measurement as a design principle. Proposition 1 gives a clean way to think about when a reasoning system may postpone its decisions. It may, exactly to the extent that its dynamics are affine over the alternatives it is entertaining. Everything a nonlinear reasoner gains by staying continuous, and everything it risks, is in the second-order term.

Limitations. These are from-scratch models on one synthetic task with a frozen random index bank, chosen precisely so that the phenomenon under study is not confounded by memorisation; Rizvi-Martel et al. [2026] give good reason to expect fine-tuned LLMs to behave differently, and nothing here transfers automatically to them. The task is reachability, where superposition has a known and provable advantage [Zhu et al., 2025b]; the arithmetic side of COCONUT’s split, where continuous reasoning *loses*, is not modelled here at all and is the obvious next test of whether the α account generalises. The Zeno analysis is a linearisation plus a numerical sweep, not a proof about the trained operator. Finally, τ and M were swept at evaluation and training together; disentangling train-time and test-time sharpness, as Butt et al. [2025] do, is left open.

What would need a cluster. Everything above runs on a laptop CPU. Three extensions would not: a branching-factor sweep wide enough to estimate the exponent relating frontier size to the discrete–continuous gap; an arithmetic-style task with a genuine computational rather than search bottleneck, to test whether the ordering inverts as it does between ProsQA and GSM8k; and porting the α sweep to a small transformer, where the retain gate would have to be introduced deliberately because the architecture does not provide it.

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